

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Report

Executive Summary

This report evaluates `openai/gpt-4o-mini` across Infron and OpenRouter under prompt-caching workloads. The run uses only `/v1/chat/completions`, keeps model/platform default reasoning and thinking behavior, and keeps prompt-length stratification enabled.

After A/B filtering, the analysis retains **783 paired samples** and **3132 request-level observations**. It excludes **34 records**. Provider model IDs are Infron `openai/gpt-4o-mini` and OpenRouter `openai/gpt-4o-mini`.

Core finding: OpenRouter leads token-level cache hit rate and observed cost in every routing mode. Throughput, E2E latency, and Streaming TTFT split 2/4 vs 2/4 across routing modes, so neither side shows a clear cross-mode lead on those performance axes.

Quality Gate

Item	Value
Model	<code>openai/gpt-4o-mini</code>
Provider model IDs	Infron <code>openai/gpt-4o-mini</code> ; OpenRouter <code>openai/gpt-4o-mini</code>
API protocol	<code>/v1/chat/completions</code> only
Groups / rounds	4 groups x 50 rounds
Prompt tiers	<code>short</code> ≈1500, <code>medium</code> ≈8000, <code>long</code> ≈32000 tokens
Reasoning / thinking	Default platform/model behavior; no explicit parameter
Pairing rule	strict sort/group/round pair with first/second <code>usage.prompt_tokens</code> deltas ≤ 50
Retained pairs	783
Request-level rows	3132
Excluded records	34

Route–Level Result Matrix

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	0	Tie 0.00% / 0.00%	Tie \$0.00000000 / \$0.00000000	Tie 0.00 tok/s / 0.00 tok/s	Tie 0.00 ms / 0.00 ms	Tie 0.00 ms / 0.00 ms
Price First	0	Tie 0.00% / 0.00%	Tie \$0.00000000 / \$0.00000000	Tie 0.00 tok/s / 0.00 tok/s	Tie 0.00 ms / 0.00 ms	Tie 0.00 ms / 0.00 ms
Latency First	0	Tie 0.00% / 0.00%	Tie \$0.00000000 / \$0.00000000	Tie 0.00 tok/s / 0.00 tok/s	Tie 0.00 ms / 0.00 ms	Tie 0.00 ms / 0.00 ms
TTFT First	0	Tie 0.00% / 0.00%	Tie \$0.00000000 / \$0.00000000	Tie 0.00 tok/s / 0.00 tok/s	Tie 0.00 ms / 0.00 ms	Tie 0.00 ms / 0.00 ms

API Protocol Record

API protocol	Endpoint	Platform	Planned pairs	Requests	Success rate	Usage coverage	Cost coverage	Cache telemetry coverage	HTTP status
chat_completions	/v1/chat/completions	Infron	0	0	0.00%	0.00%	0.00%	0.00%	
chat_completions	/v1/chat/completions	OpenRouter	0	0	0.00%	0.00%	0.00%	0.00%	

Prompt–Length Stratified Results

Tier	Pairs	Token cache hit rate	Observed cost	E2E latency	TTFT
short	0	Infron 91.85% / OpenRouter 92.21%	Infron \$0.11666700 / OpenRouter \$0.08010330	Infron 2083.56 ms / OpenRouter 2081.97 ms	Infron 1744.61 ms / OpenRouter 1844.16 ms
medium	0	Infron 97.52% / OpenRouter 99.03%	Infron \$0.59036700 / OpenRouter \$0.37255890	Infron 2737.27 ms / OpenRouter 2803.95 ms	Infron 2389.97 ms / OpenRouter 2544.34 ms
long	0	Infron 98.14% / OpenRouter 98.91%	Infron \$2.29457600 / OpenRouter \$1.44044130	Infron 3526.13 ms / OpenRouter 3412.58 ms	Infron 3111.83 ms / OpenRouter 3123.59 ms

Reasoning / Thinking Observation

This run does not explicitly disable or force reasoning/thinking. Reasoning tokens are treated as observed telemetry under model and platform defaults.

Reproducibility References

Artifact	Link	SHA256 / Notes
Chinese HTML report	< https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/reports/routing-cache-cost-streaming-performance-ab-study__gpt-4o-mini__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-06.zh.html >	
English HTML report	< https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/reports/routing-cache-cost-streaming-performance-ab-study__gpt-4o-mini__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-06.en.html >	
Summary	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/summary.json >	906b4557b6b175f514bd7a663bc404f68f4c6e86c8078df6861bc907e
Paired dataset CSV	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/benchmark_pairs.csv >	a04ef66ecb5882f82aa039b2dabc41c4d2e6a0f36bedf88c542b5c06bb
Request-level dataset JSONL	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/benchmark_requests.jsonl >	853de5f12f65c3217531b5caedb8d1b94238521baec04908e318bf167

Artifact	Link	SHA256 / Notes
Excluded-record audit	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/records_excluded.json >	7c67d0f35c7a36a58703d6190af49ad14ba32379a6dc872a9739d3330
Data directory	< https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data >	
Benchmark runner source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/rerun_routing_sort_cache_cost_ab.py >	b8ff71395fb08a6ff817c03d153ac09914b2bddb6994265a86a5ecaba
HTML report renderer source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/render_glm52_deepseek_style_report.py >	411289b9087ee50275a2d08da86cb7342a9c81933764c05fe08acd684
Test source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/test_rerun_routing_sort_cache_cost_ab.py >	30b46e2b21db1b0e42899db12983d8e5bdfbaa8ceb9c64b04e5e664aa
A/B report standard	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-4o-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/ab-report-standard.md >	81c99b29dceaf28a80ba9a12d648e138f7aa21b37b35c109f7ae30145

Dataset reference: `business_representative` built-in representative business templates; exported request-level dataset is `benchmark_requests.jsonl`.